DIGITAL TWIN CHALLENGE 2026 · PROTOTYPE & CODE SUBMISSION

VITALTWIN

“Your health. Your baseline. Your digital twin.”

An intelligent patient Digital Twin prototype that learns a personal baseline, observes changing health signals, detects meaningful deviations, and explains what it sees.

TEAM

TEAM	After Eclipse
TEAM LEADER	Akash Selvaraj R
COLLEGE	SRM Institute of Science and Technology

MIT LICENSE · OPEN SOURCE



SYNTHETIC / SIMULATED PROTOTYPE DATA

A living, personalized representation of a patient's physiological state — non-diagnostic, browser-local, and fully explainable.

github.com/Akash-Selvaraj-R/VitalTwin

HEALTHCARE IS OFTEN REACTIVE

A patient's current condition is too often interpreted without enough personalized context — and generic reference ranges can miss what is unusual for a specific individual.

PATIENT DATA

Heart rate. Oxygen. Temperature.
More numbers than anyone can interpret.

FRAGMENTED CONTEXT

Each value read alone, against a population average — not against the patient.

REACTIVE INTERPRETATION

Change is noticed only after it becomes symptomatic, urgent, or alarming.

“What if healthcare could understand change relative to the patient's own baseline?”

A heart rate of 88 bpm can be unremarkable for one person and a clear departure from their own pattern for another. Normal is personal — but almost nothing is measured against it.

MEET VITALTWIN

“VitalTwin is a patient-focused Digital Twin prototype that creates a living, personalized representation of a patient's physiological state.”

01 PROFILE

The user provides basic profile information — age, sex, body measures, activity level.

02 BASELINE

The system establishes a personal baseline: one normal value per signal, for this person.

03 SIMULATION

Simulated physiological signals evolve over time, deterministically and repeatably.

04 DEVIATION

The twin interprets movement away from that personal baseline — not a generic range.

05 INTELLIGENCE

Meaningful, sustained change becomes a calm, timestamped intelligence event.

06 WHY

The patient can always see exactly why the twin changed state.

ONE JOURNEY, START TO FINISH

Every step below is implemented in the current prototype — nothing here is conceptual.

1. **MEET VITALTWIN**
   Boot splash and a time-aware greeting.
2. **ENTER BASIC PROFILE**
   Onboarding with validation, stored locally.
3. **ESTABLISH BASELINE**
   Personal baseline derived from the profile.
4. **GUIDED 7-STEP TOUR**
   A replayable walkthrough of the whole UI.
5. **START LIVE MONITORING**
   Signals begin evolving; the twin comes alive.
6. **OBSERVE TWIN STATE**
   STABLE · ATTENTION · RECOVERING, with hysteresis.
7. **UNDERSTAND INTELLIGENCE**
   Contextual events, each fully explainable.
8. **EXPLORE HISTORY**
   24H / 7D / 30D timeline plus live captures.
9. **MANAGE PROFILE & SETTINGS**
   Edit profile, reduced motion, logout, clear data.

A TWIN THAT REACTS, IN REAL TIME

LIVE MONITORING

A 1200 ms tick loop drives every signal on screen.

SIGNAL CARDS ×6

HR, SpO_2, temperature, respiratory rate, stress load, recovery — each vs. its baseline.

HEALTH SCORE

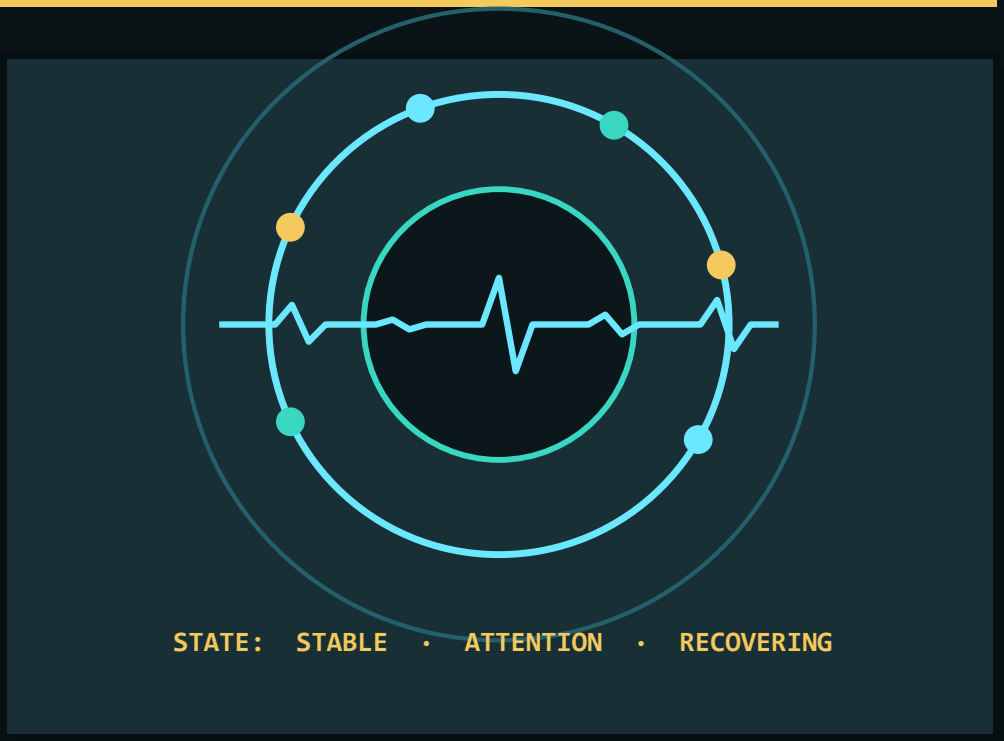
A transparent blend of alignment, recovery and stability — never clinical.

TWIN STATE

The visualization reacts to STABLE · ATTENTION · RECOVERING.

CONTEXTUAL INTELLIGENCE

Only meaningful transitions generate an event — never noise.



SYNTHETIC / SIMULATED PROTOTYPE DATA

No medical measurement is taken. The visualization is an abstract, state-reactive representation of simulated signals.

YOUR NORMAL IS PERSONAL

GENERIC THRESHOLDS ASK

“Is this value normal?”

VITALTWIN ASKS

“Is this meaningfully different from THIS patient's baseline?”

Baselines are derived per signal from age, sex, body measures (BMI) and activity level — then every reading is judged against that person's own pattern. This enables patient-specific interpretation instead of one universal cut-off.

PERSONAL BASELINE

derived at onboarding

▼

CURRENT SIGNAL

simulated reading this tick

▼

DEVIATION

% and direction vs. baseline

▼

SUSTAINED CHANGE

held across consecutive ticks

▼

TWIN STATE

STABLE · ATTENTION · RECOVERING

06 · THE ENGINE

SIGNALS TO INTELLIGENCE, EVERY TICK



THE ONLY THREE TWIN STATES

STABLE

No sustained deviation beyond personal tolerance.

ATTENTION

≥1 major, or ≥3 sustained off-baseline signals.

RECOVERING

Deviations shrinking; settles back toward STABLE.

DEMO SCENARIOS — INPUTS, NOT STATES

STABLE DAY · ELEVATED SIGNALS · RECOVERY steer the simulation from the outside. The patient-facing state is always STABLE, ATTENTION or RECOVERING.

REPRODUCIBLE BY DESIGN

`ACTIVATION_TICKS = 5 · tick = 1200 ms · score weights 0.42 / 0.33 / 0.25`

Current prototype uses a deterministic rule / state engine — not a trained ML model. No training data, no inference runtime, no Math.random.

EVERY ALERT EXPLAINS ITSELF

Every intelligence event opens into the same six-step causal chain — in plain, patient-friendly language.

1. WHAT THE TWIN SAW
the observed simulated reading

2. YOUR BASELINE
the personal value it was compared against

3. CHANGE DETECTED
size and direction of the deviation

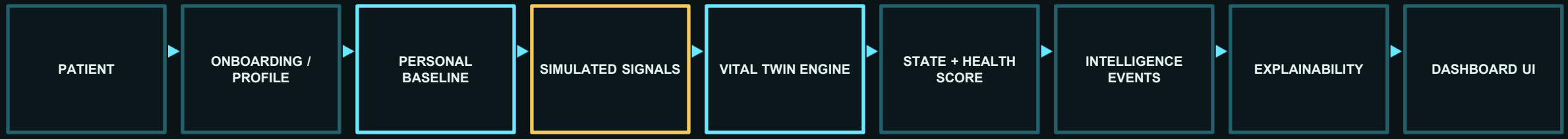
4. SIGNALS INVOLVED
which signals moved together

5. WHY IT MATTERS
context in careful, non-diagnostic language

6. WHAT YOU CAN DO
safe, general next-step considerations

“Explainability is part of the experience, not an afterthought.” No opaque alerts, no black-box scores — and never a diagnosis.

BUILT CLIENT-SIDE, END TO END



NEXT.JS 16 · REACT 19 · TYPESCRIPT 5.7

app/page.tsx composition root · app/layout.tsx · Tailwind CSS 4 neo-brutalist design system

CORE LOGIC — NO REACT, NO DOM

lib/engine.ts — pure deterministic simulation, baselines, state machine, scoring, events, history

REACT BINDING

lib/use-vital-twin.ts — tick loop with cleanup, scenario controls, overlays, keyboard a11y

12 REACT COMPONENTS

Onboarding · DigitalTwin + EcgStrip · MonitoringStrip · SignalCard ×6 · HealthScore · TwinSees · IntelligenceFeed · ExplainabilityDrawer · HistoryDrawer · AccountDrawers · TourOverlay · DemoControls

PERSISTENCE BOUNDARY

Single-browser prototype storage:

`localStorage · vital-twin-profile`
`localStorage · vital-twin-reduced`

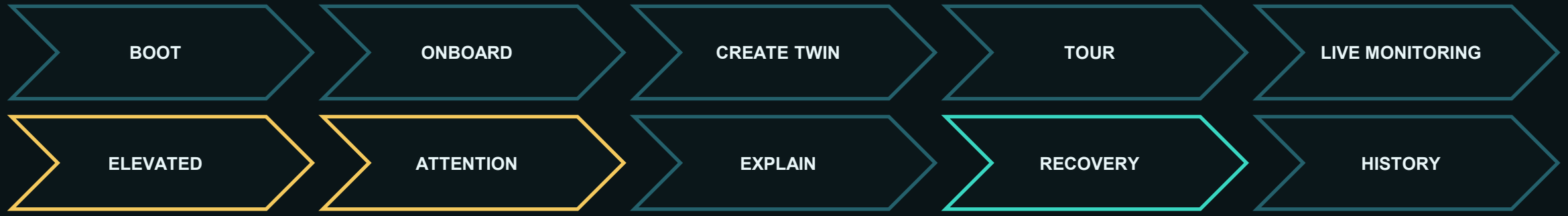
Sanitized on read. Nothing leaves the device.

PROTOTYPE BOUNDARY

No backend · no cloud patient record · no wearable feed · no EHR · no ABHA · no real-time clinical data.

WHAT YOU WILL SEE

A complete, repeatable walkthrough — every state can be reached, explained, and exited. No dead ends.



DEMO CONTROLS — BUILT IN

+1 HOUR	fast-forward one simulated hour
+6 HOURS	fast-forward a full evening
PAUSE	freeze the simulation mid-flight
RESET	return to the personal baseline

SCENARIO SWITCHES

STABLE DAY · ELEVATED SIGNALS · RECOVERY

Watch a single change propagate through the entire product: scenario → signals → deviation from the personal baseline → twin state → intelligence event → explainability → recovery → history.

RADICALLY HONEST ABOUT SCOPE

WHAT THIS PROTOTYPE IS

- Synthetic / simulated physiological signals, generated on-device
- Browser-local profile persistence (localStorage), sanitized on read
- Non-diagnostic language throughout — signals, patterns, baselines
- A research / competition proof-of-concept, MIT licensed

WHAT THIS PROTOTYPE IS NOT

- No real patient records — no real patient data is used or required
- No diagnosis, treatment, or emergency response
- No medical-device, regulatory, or clinical-validation claim
- No ABHA · no EHR · no wearable · no backend patient database

Designed as a foundation for future real-world integration.

Wearables, longitudinal patient data, cloud sync, ABHA/ABDM and clinically validated models are future scope — clearly separated from what ships today.

11 · CLOSING

FROM GENERIC THRESHOLDS TO PERSONAL CONTEXT.

VitalTwin turns a patient's baseline into an interactive digital health experience.

DIGITAL TWIN + PERSONAL BASELINE + CONTINUOUS SIMULATION + EXPLAINABLE INTELLIGENCE

= PROACTIVE HEALTH INTELLIGENCE

TEAM AFTER ECLIPSE
Akash Selvaraj R · SRM Institute of Science and Technology
Digital Twin Challenge 2026

github.com/Akash-Selvaraj-R/VitalTwin
Simulated data · Non-diagnostic prototype
Not a medical device